

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1– Concept Mapping

Course Code:	CSA0302	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – Concept Mapping
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
Student Name:	Rieona J	Reg. No.:	192424268
Section / Batch:	AIDS	Date:	25-07-2026

Code of Conduct

I (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Instructions

- This test contains 2 concept mapping, Total: 100 Marks.
- Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification , Linkages, Organization, Integration of Literature, and Clarity
- No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

Rubrics

Questions	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Total Marks
Q1						
Q2						

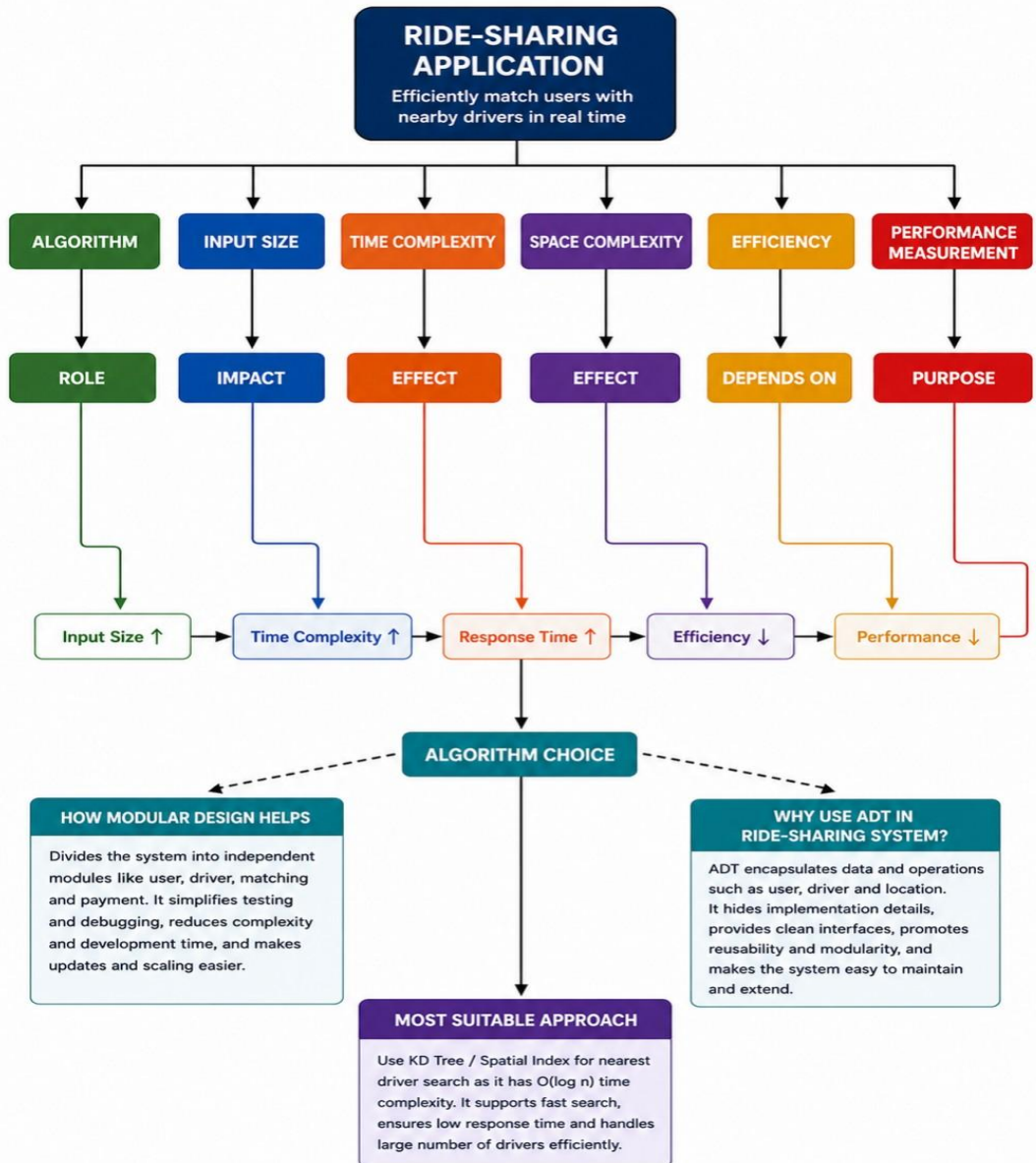
Annexure A – Concept Mapping

Questions:

EACH QUESTIONS CARRIES : 10 MARKS

Duration :ONE HOUR

Q4 A ride-sharing application must efficiently match users with nearby drivers in real time. Construct a concept map connecting Algorithm → Input Size → Time Complexity → Space Complexity → Efficiency → Performance Measurement. Illustrate how increasing input size impacts system performance and establish relationships between these concepts. Analyze how the choice of algorithm affects efficiency and justify the most suitable approach.



Q14 A ticket booking system supports operations such as booking, cancellation, and seat allocation. Construct a concept map connecting Abstract Data Type (ADT) → Operations → Implementation → Data Structures → Performance. Analyze how different implementations (array vs linked list) affect efficiency. Justify the best choice for real-time booking systems.

